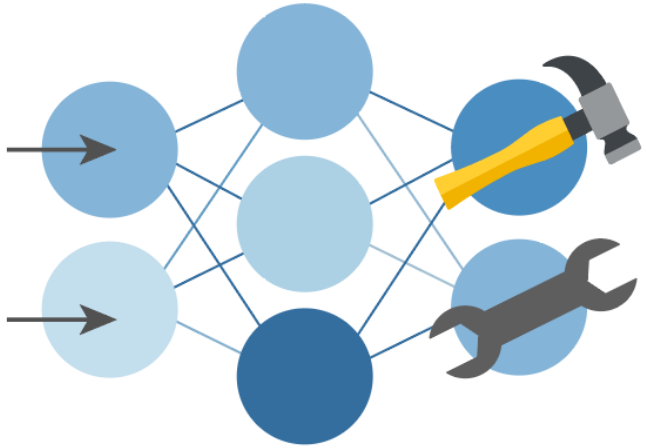
Python + Agents



🛠️ Feb 24: Building your first agent in Python
🧠 Feb 25: Adding context and memory to agents
📊 Feb 26: Monitoring and evaluating agents
🔀 Mar 3: Building your first AI-driven workflows
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Microsoft

Python + Agents

🛠️ Building your first agent in Python

aka.ms/pythonagents/slides/building

Pamela Fox

Python Cloud Advocate

www.pamelafox.org

Today we'll cover...

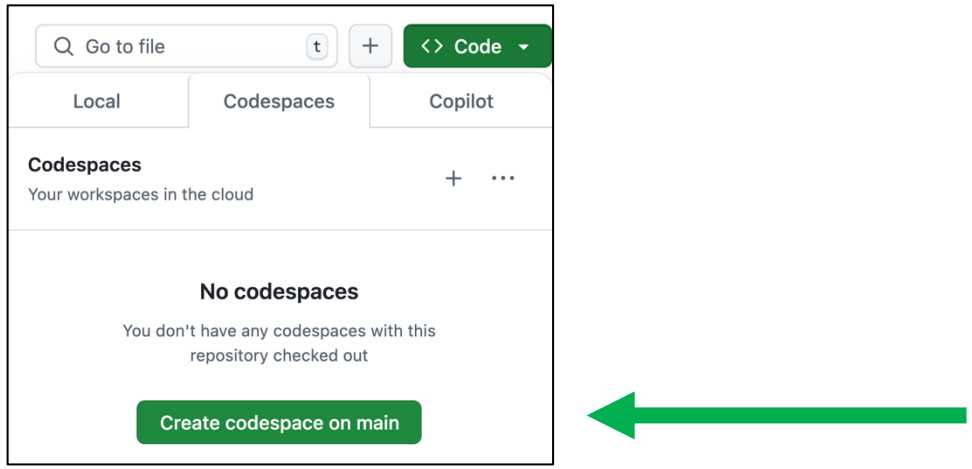
- What's an agent?
- How tool-calling works
- Building agents with agent-framework
- Integrating agents with MCP servers tools
- Agent middleware
- Supervisor agent architecture

Want to follow along?

1. Open this GitHub repository:

aka.ms/python-agentframework-demos

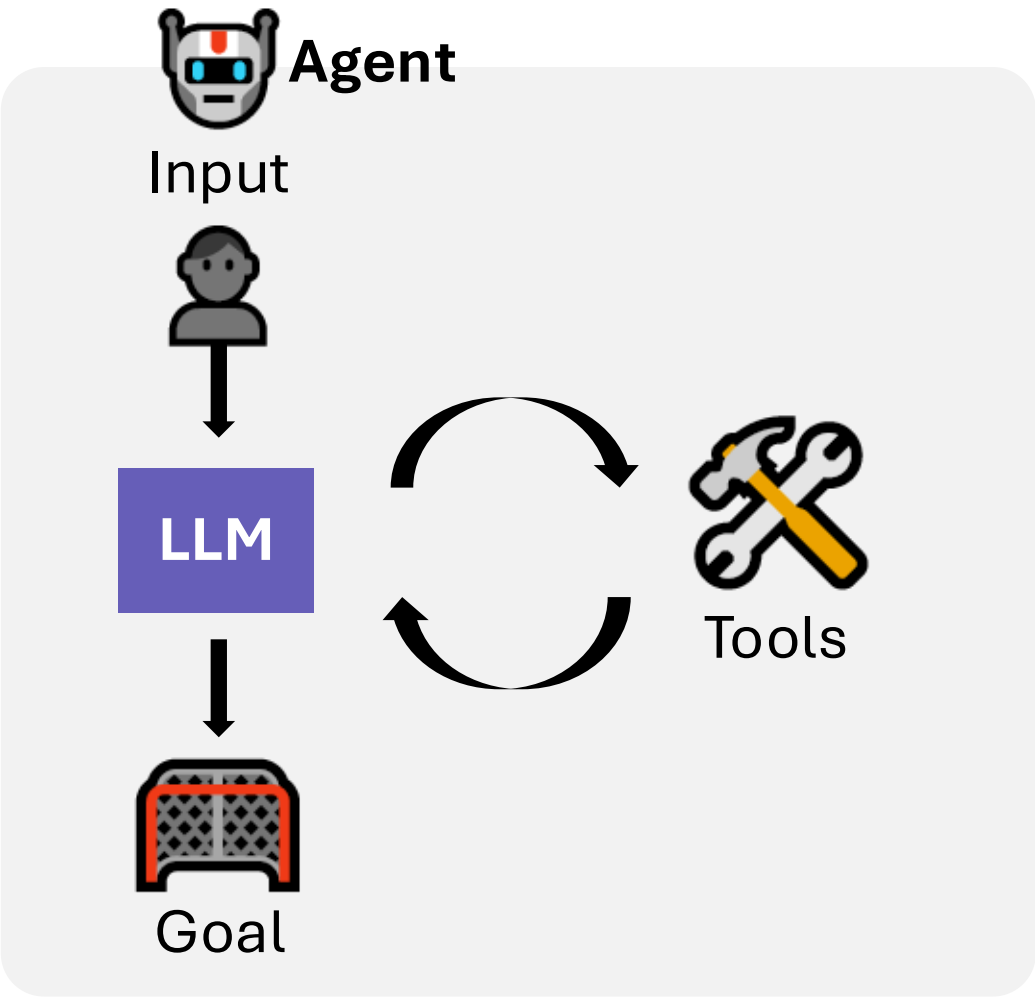
2. Use "Code" button to create a GitHub Codespace:



3. Wait a few minutes for Codespace to start up ⏱️

Agents 101

What is an agent?



An **AI agent** uses an **LLM** to run **tools** in a loop to achieve a **goal**.

Agents are often augmented by:

- Context
- Memory
- Planning
- Humans

https://simonwillison.net/2025/Sep/18/agents/

Python AI agent frameworks

Framework	Description
agent-framework	A framework from Microsoft with support for agentic patterns and full integration with Azure offerings. (Successor to autogen and semantic-kernel)
langchain v1	An agent-centric framework built on top of Langgraph, with optional Langsmith monitoring
pydantic-ai	A flexible framework designed for type safety and observability (Logfire/OTel), from the creators of Pydantic
openai-agents	A framework from OpenAI that is optimized for use with OpenAI models (especially Responses API/reasoning).

Calling tools without a framework

What is "tool calling"?

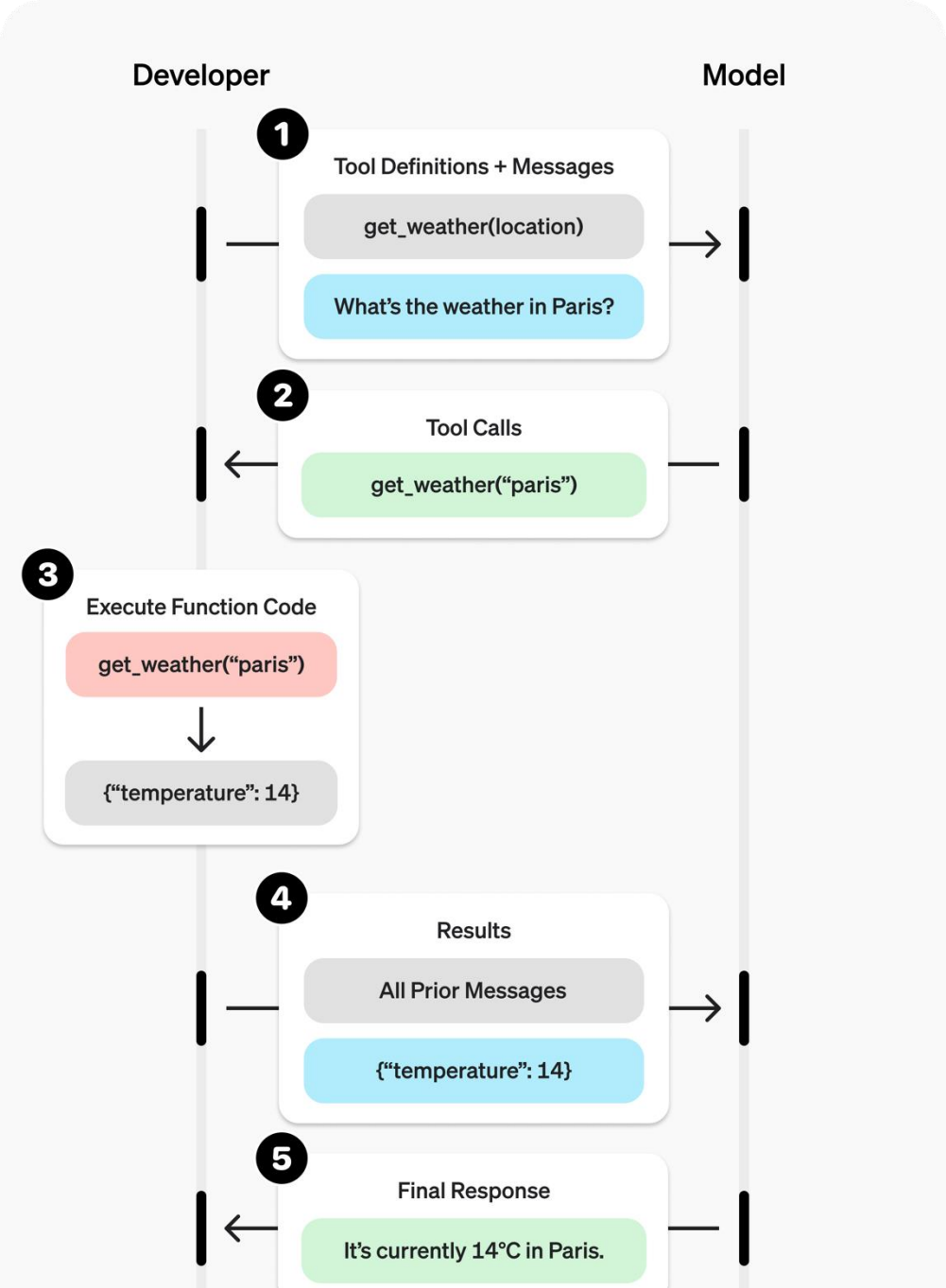
**Also known as "function calling"*

- Tool calling permits an LLM to not just generate text, but also generate **function calls** based off function definitions
- The LLM responds with the **function name** and **the arguments** necessary to satisfy the request
- This makes it easy to connect LLMs with **APIs**, **databases**, and arbitrary **Python logic**.

https://platform.openai.com/docs/guides/function-calling

Tool calling flow

1. The code tells LLM what tools they can call
2. The LLM responds with suggested tool name and arguments
3. The code calls function for that tool
4. The code sends prior messages and return value from tool function to LLM
5. The LLM responds based off full history



https://platform.openai.com/docs/guides/function-calling

Step 1) Tell the LLM what functions it can call

```
tools = [{
  "type": "function",
  "function": {
    "name": "lookup_weather",
    "description": "Lookup the weather for
a given city name or zip code.",
    "parameters": {
      "type": "object",
      "properties": {
        "city_name": {
          "type": "string",
          "description": "The city name",
        },
        "zip_code": {"type": "string",
          "description": "The zip code"
}}}}}]
```

```
response = client.chat.completions.create(
  model="gpt-5",
  messages=[
   {"role": "system",
    "content": "You're a weather chatbot."},
   {"role": "user",
    "content": "is it sunny in LA, CA?"},
  ],
  tools=tools,
)
```

[Full example: openai_tool_calling.py]

Step 2) Get function name and args from response

```
if response.choices[0].message.tool_calls:
  tool_call = response.choices[0].message.tool_calls[0]
  function_name = tool_call.function.name
  function_arguments = json.loads(tool_call.function.arguments)
  print(function_name)
  print(function_arguments)
else:
  print(response.choices[0].message.content)
```

```
lookup_weather
{"city_name":"Los Angeles"}
```

Step 3) Call a local function based on response

```
def lookup_weather(city_name=None, zip_code=None):
  print(f"Looking up weather for {city_name or zip_code}...")
  return {"city_name": city_name, "zip_code": zip_code,
          "weather": "sunny", "temperature": 75}

# ..Make call to the LLM here...

if response.choices[0].message.tool_calls:
  tool_call = response.choices[0].message.tool_calls[0]
  function_name = tool_call.function.name
  function_arguments = json.loads(tool_call.function.arguments)
  if function_name == "lookup_weather":
    lookup_weather(**function_arguments)
```

```
Looking up weather for Los Angeles...
```

Step 4) Send tool results to LLM for final answer

```
if response.choices[0].message.tool_calls:
  tool_call = response.choices[0].message.tool_calls[0]
  function_name = tool_call.function.name
  function_arguments = json.loads(tool_call.function.arguments)

  if function_name == "lookup_weather":
    messages.append(response.choices[0].message)
    result = lookup_weather(**function_arguments)
    messages.append({
      "role": "tool", "tool_call_id": tool_call.id, "content": json.dumps(result)})
    response = client.chat.completions.create(
      model="gpt-5", messages=messages, tools=tools)
    print(response.choices[0].message.content)
```

```
Yes — it's currently sunny in Los Angeles, CA with a temperature around 75°F.
```

Building agents with agent-framework

https://learn.microsoft.com/agent-framework/

Install agent-framework

We recommend installing the subpackages for agent-framework instead of the entire agent-framework package, to reduce indirect dependencies.

Today's examples require:

```
agent-framework-core
agent-framework-devui
```

Our repo installs from a very recent git commit, since MAF had recent big changes:

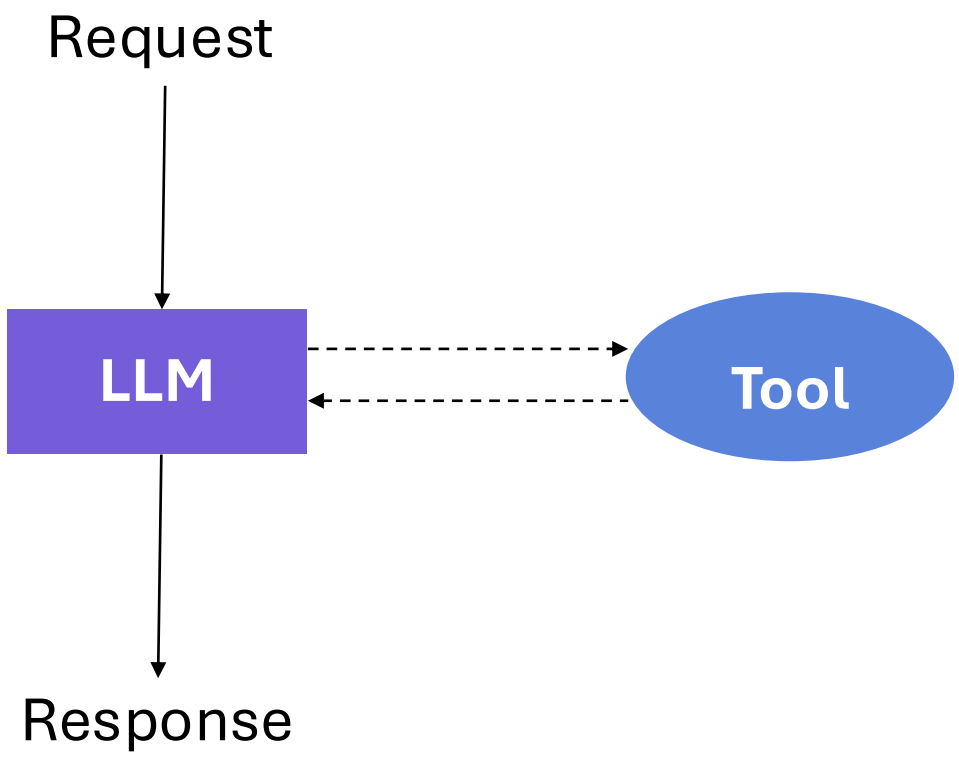
```
agent-framework-core @ git+https://github.com/microsoft/agent-framework.git@11628c3166a1845683c5aef1e0d389eb862bcbaa#subdirectory=python/packages/core
```

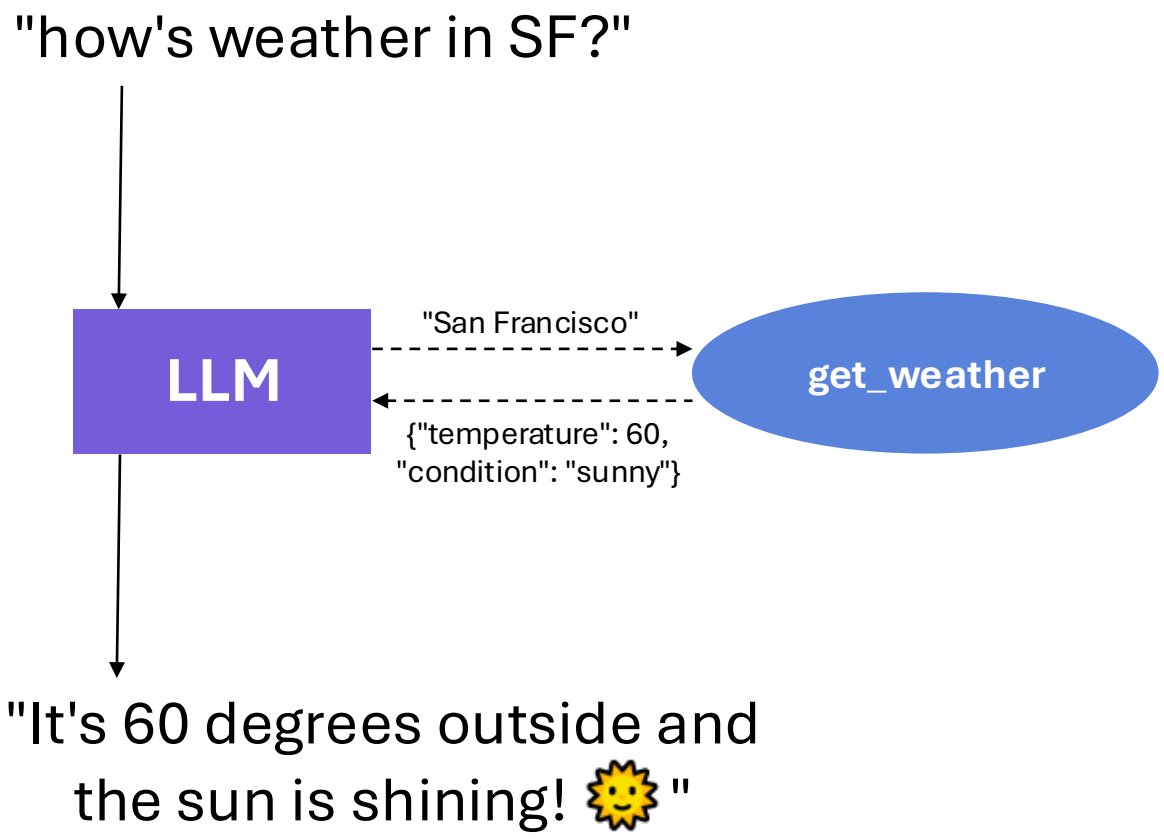
Full requirements: [pyproject.toml]

Agent architecture:

Single agent with tool

The agent can use a tool to help answer a user's question.





agent-framework

Single agent with tool

Define tools using Python functions, with typed args, return type, and docstring:

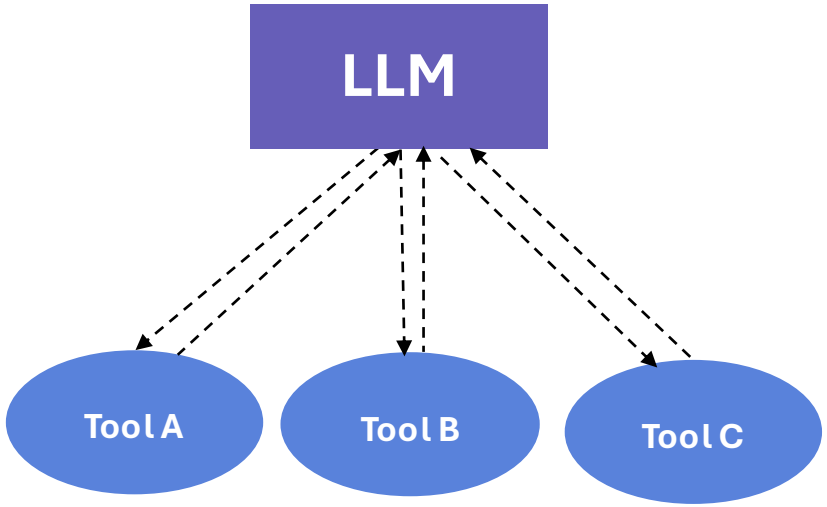
```
@tool
def get_weather(
  city: Annotated[str, Field(description="City name, spelled out fully")],
  ) -> dict:
  """Returns weather data, a dict with temperature and description."""
  return {"temperature": 72, "description": "Sunny"}

agent = Agent(client=client,
  instructions="You're an info agent. Answer questions cheerfully.",
  tools=[get_weather])
response = await agent.run("Whats weather today in San Francisco?")
print(response.text)
```

Full example: [agent_tool.py]

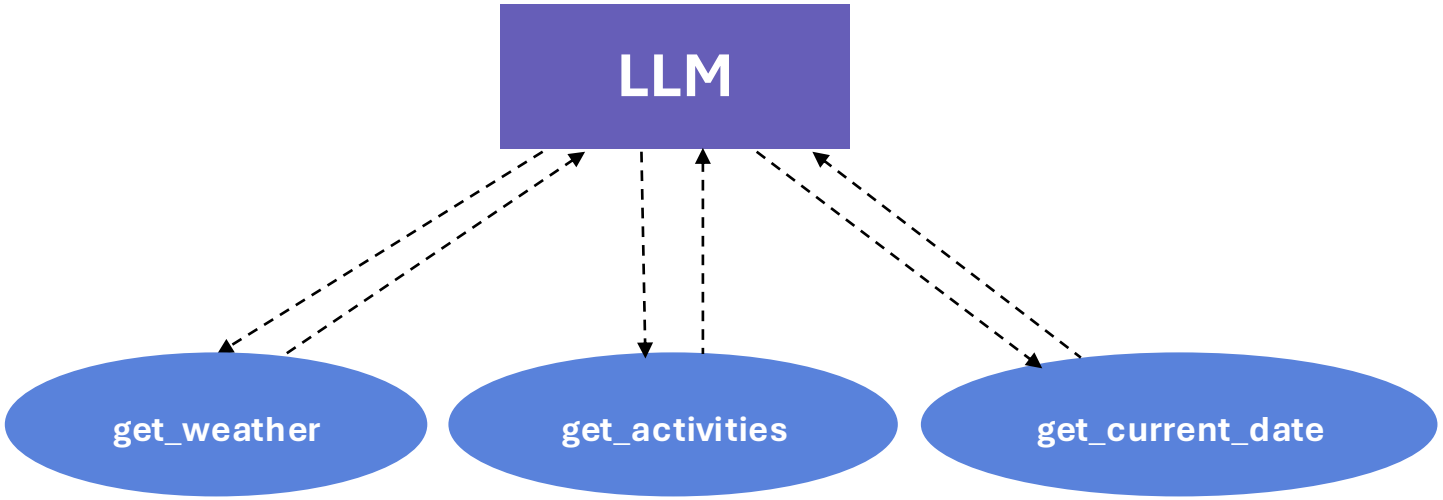
Agent architecture:

Single agent with multiple tools

The agent must decide which tool to use and what order to call tools in.



Weekend planner agent



agent-framework

Single agent with tools

```
@tool
def get_current_date() -> str:
  return datetime.now().strftime("%Y-%m-%d")

@tool
def get_weather(
  city: Annotated[str, Field(description="The city to get weather for")]) -> dict:
  ...

@tool
def get_activities(
  city: Annotated[str, Field(description="The city to get activities for")],
  date: Annotated[str, Field(description="Date, in format YYYY-MM-DD.")]) -> list:
  ...

agent = Agent(client=client, instructions="You help users plan their weekends.",
  tools=[get_weather, get_activities, get_current_date])
```

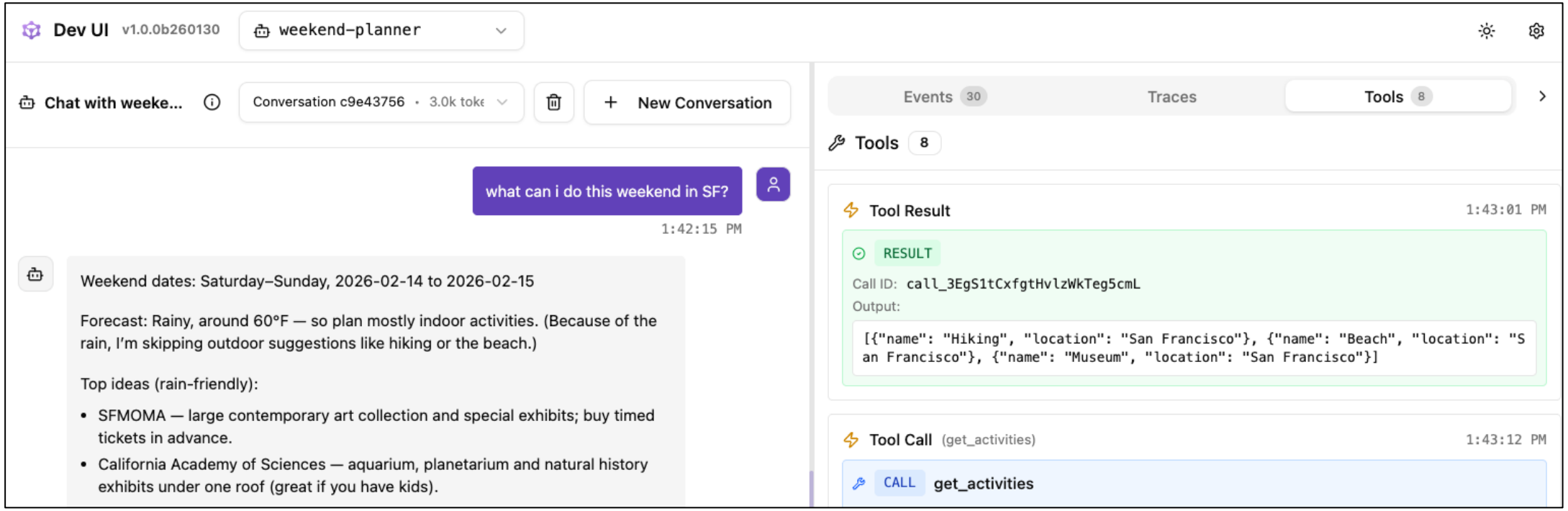
Full example: [agent_tools.py]

agent-framework

Local experimentation with DevUI

DevUI makes it easy to chat with your agents and trace tool calls:

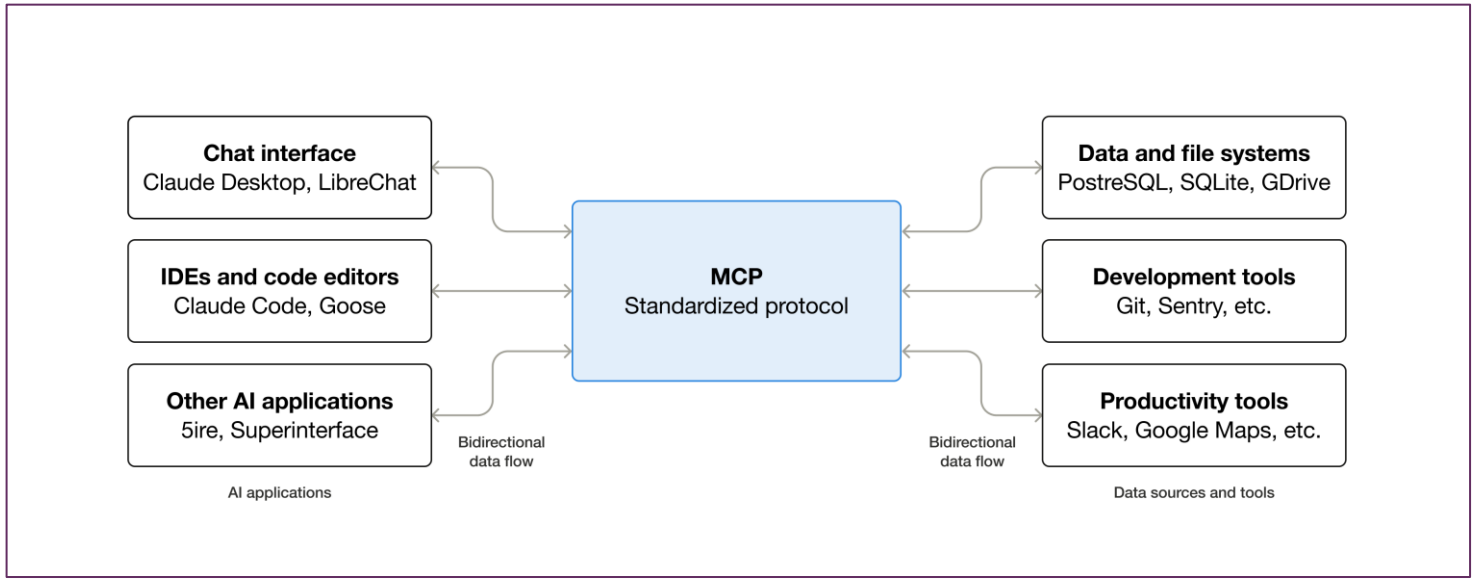
```
from agent_framework.devui import serve

serve(entities=[agent], auto_open=True)
```

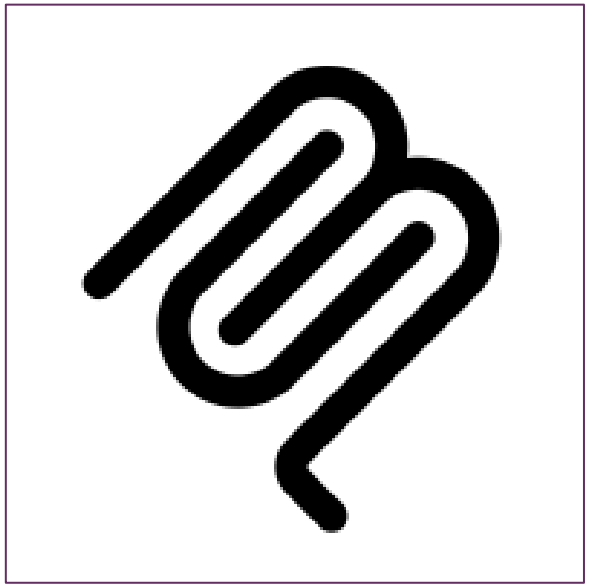


Integrating agents with MCP server tools

Model Context Protocol

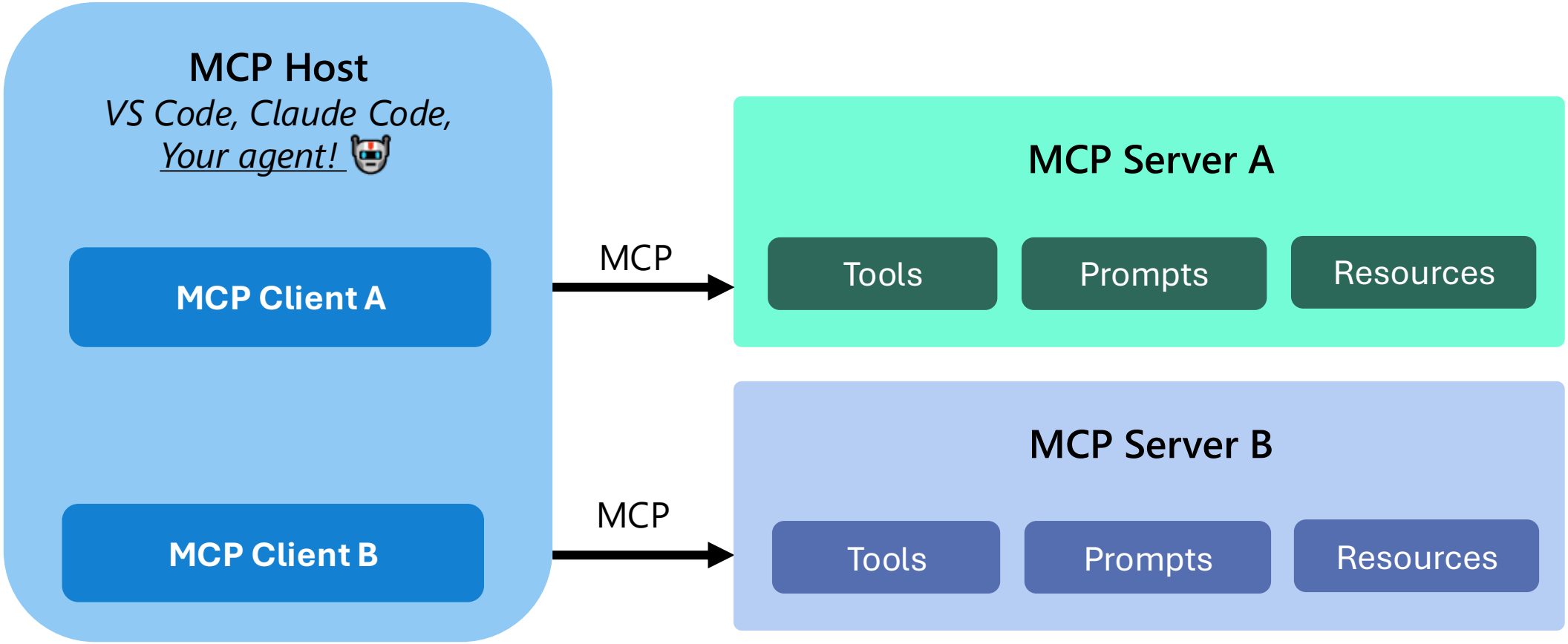
An **open protocol** created by Anthropic that **standardizes** the interaction between LLMs and **external tools**, **data sources**, and **applications**.





https://modelcontextprotocol.io/

MCP client + server architecture

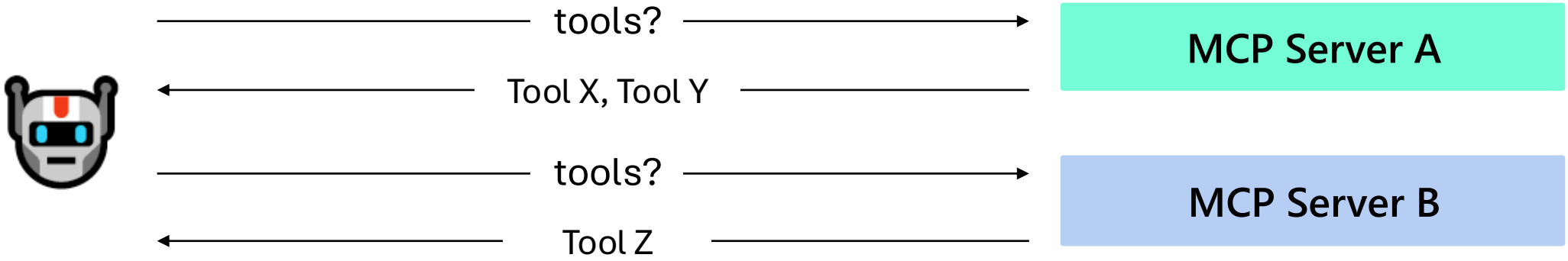


https://modelcontextprotocol.io/docs/learn/architecture

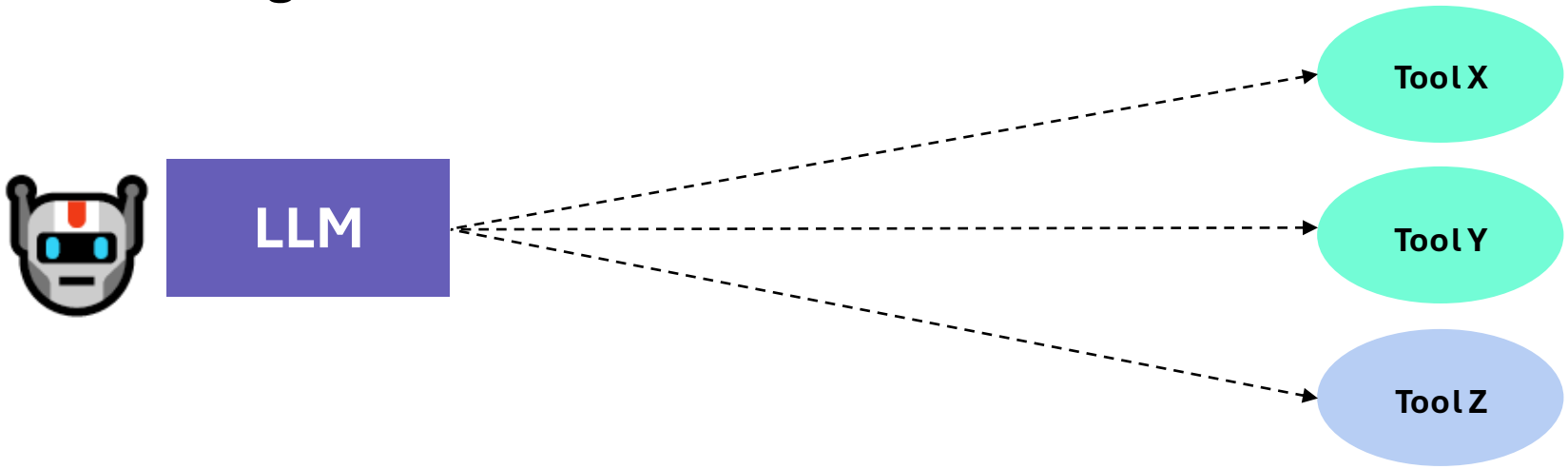
Agent architecture:

Single agent with MCP servers

First the agent queries each MCP server to discover their tools.



Then the agent decides which of the MCP tools to call.



agent-framework

Agent with local MCP server

First run the server:

```
uv run examples/mcp_server.py
```

Then run the agent:

```
async with (
  MCPStreamableHTTPTool(name="Local MCP Server",
     url="http://localhost:8000/mcp") as mcp_server,
  Agent(client=client,
    instructions=f"You help users with tasks using the available tools.
                   Today's date is {datetime.now().strftime('%Y-%m-%d')}.",
    tools=[mcp_server]) as agent,
  ):
    response = await agent.run("I bought laptop today for $1200 on my visa.")
```

Full example: [agent_mcp_local.py]

agent-framework

Agent with remote MCP server

```
async with (
  MCPStreamableHTTPTool(
    name="Microsoft Learn MCP",
    url="https://learn.microsoft.com/api/mcp") as mcp_server,
  Agent(
    client=client,
    instructions="You help with Microsoft documentation questions. Use the
available tools to search for relevant docs.",
    tools=[mcp_server]) as agent,
  ):
    response = await agent.run(
       "How do I create an Azure Blob storage account using the az cli?")
```

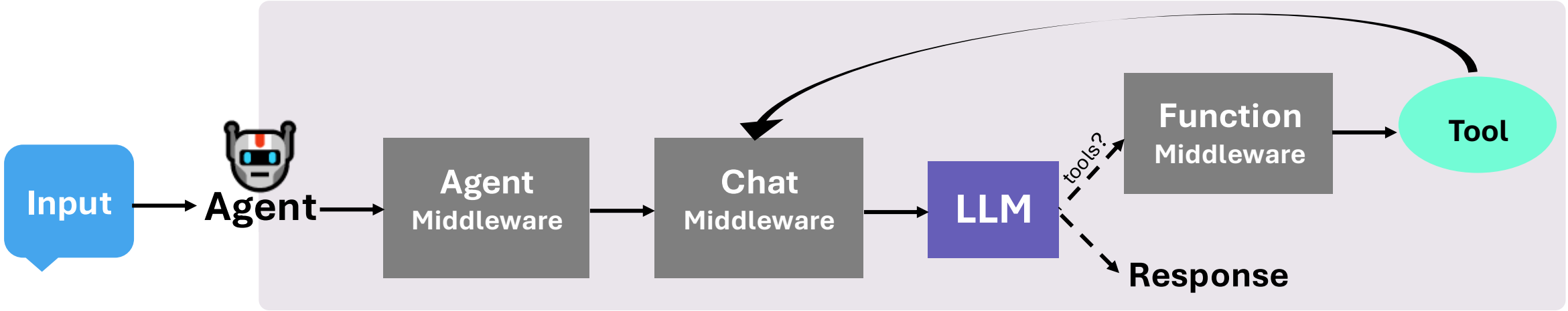
Full example: [agent_mcp_remote.py]

Agent middleware

Agent architecture:

Agent middleware

Middleware can be called at different stages in the process of running an agent:



Each middleware receives a context:

AgentContext:
- **messages**
- **is_streaming**
- **metadata**
- **agent**
- **result:** mutate this to change agent response
- **terminate**: set this flag to stop processing

ChatContext:
- **messages**
- **is_streaming**
- **metadata**
- **chat_client**
- **chat_options**
- **result:** mutate this to change chat response
- **terminate**: set this flag to stop processing

FunctionInvocationContext:
- **metadata**
- **function**
- **arguments**
- **result:** mutate this to change the return value
- **terminate**: set this flag to stop processing

agent-framework

Agent Middleware example

A middleware should process, `call_next()`, then optionally post-process:

```
async def timing_agent_middleware(
  context: AgentContext,
  call_next: Callable[[], Awaitable[None]],
  ) -> None:

  start = time.perf_counter()
  logger.info("[🕒 Timing Agent Middleware] Starting agent execution")

  await call_next()

  elapsed = time.perf_counter() - start
  logger.info(f"[🕒 Timing Agent Middleware] Execution took {elapsed:.2f}s")
```

Full example: [agent_middleware.py]

agent-framework

Chat Middleware example

A middleware can access the `context` for more details:

```
async def logging_chat_middleware(
  context: ChatContext,
  call_next: Callable[[], Awaitable[None]],
  ) -> None:

  logger.info(f"Sending {len(context.messages)} messages to AI")

  await call_next()

  logger.info("AI response received")
```

Full example: [agent_middleware.py]

agent-framework

Function Middleware example

The `context` may change after `next(context)` is called:

```
async def logging_function_middleware(
  context: FunctionInvocationContext,
  call_next: Callable[[], Awaitable[None]]
  ) -> None:

  logger.info(f"Calling {context.function.name} with: {context.arguments}")

  await call_next()

  logger.info(f"Function {context.function.name} returned: {context.result}")
```

Full example: [agent_middleware.py]

agent-framework

Class-based middleware

A middleware can be defined as a subclass with a `process()` method:

```
class MessageCountChatMiddleware(ChatMiddleware):

  def __init__(self) -> None:
    self.total_messages = 0

  async def process(self,
    context: ChatContext,
    call_next: Callable[[], Awaitable[None]]) -> None:
    self.total_messages += len(context.messages)
    logger.info(f"Messages in this request: {len(context.messages)},
                total so far: {self.total_messages}")
    await call_next()
```

Full example: [agent_middleware.py]

agent-framework

Middleware with termination

A middleware can set `terminate()` to stop execution of current agent run:

```
class BlockingAgentMiddleware(AgentMiddleware):
  async def process(self, context: AgentContext,
      call_next: Callable[[], Awaitable[None]]) -> None:
    last_message = context.messages[-1] if context.messages else None
    for word in self.blocked_words:
      if word.lower() in last_message.text.lower():
        context.terminate = True
        context.result = AgentResponse(
            messages=[Message(role="assistant",
                      text=f"I can't process requests about '{word}'.")])
        return
    await call_next()
```

Full example: [agent_middleware.py]

agent-framework

Registering middleware per agent

Register middleware functions and class instances on the ChatAgent:

```
blocking_middleware = BlockingAgentMiddleware(blocked_words=["nuclear"])
timing_function_middleware = TimingFunctionMiddleware()
message_count_middleware = MessageCountChatMiddleware()

agent = Agent(
  client=client,
  instructions="Help users plan weekends.",
  tools=[get_weather, get_current_date],
  middleware=[
    timing_agent_middleware, blocking_middleware,
    logging_function_middleware, timing_function_middleware,
    logging_chat_middleware, message_count_middleware])
```

Full example: [agent_middleware.py]

agent-framework

Registering middleware per run

You can also register middleware for a particular run of an agent:

```
async def extra_agent_middleware(
    context: AgentContext,
    call_next: Callable[[], Awaitable[None]]) -> None:

  logger.info("This middleware only applies to this run")
  await call_next()
  logger.info("Run completed")

response = await agent.run("What's the weather like in Portland?",
  middleware=[extra_agent_middleware])
```

Full example: [agent_middleware.py]

Middleware scenarios

Agent Middleware:
- Logging & timing
- Content filtering / blocking
- Authorization checks
- Result caching
- Result override
- Thread summarization
- Dynamic system prompts

Chat Middleware:
- Token tracking
- Latency tracking
- Budget enforcement
- PII redaction
- Response moderation
- Model fallback
- Model call limits

Function Middleware:
- Audit logging
- Permission checks
- Input validation
- Caching results
- HITL approval
- Tool call limits
- Tool retry with backoff

Once you build a middleware for one agent, you can reuse across other agents. You might create a central package of middleware for many agents to use.

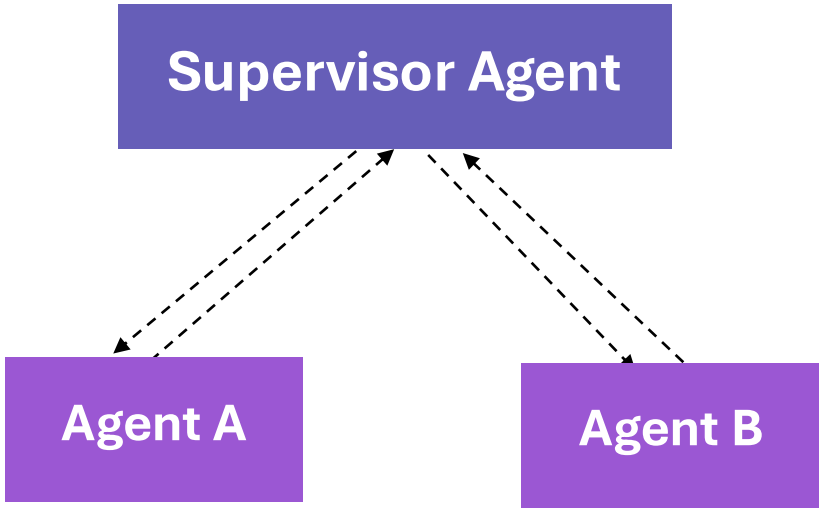
Multi-agent architectures

Agent architecture:

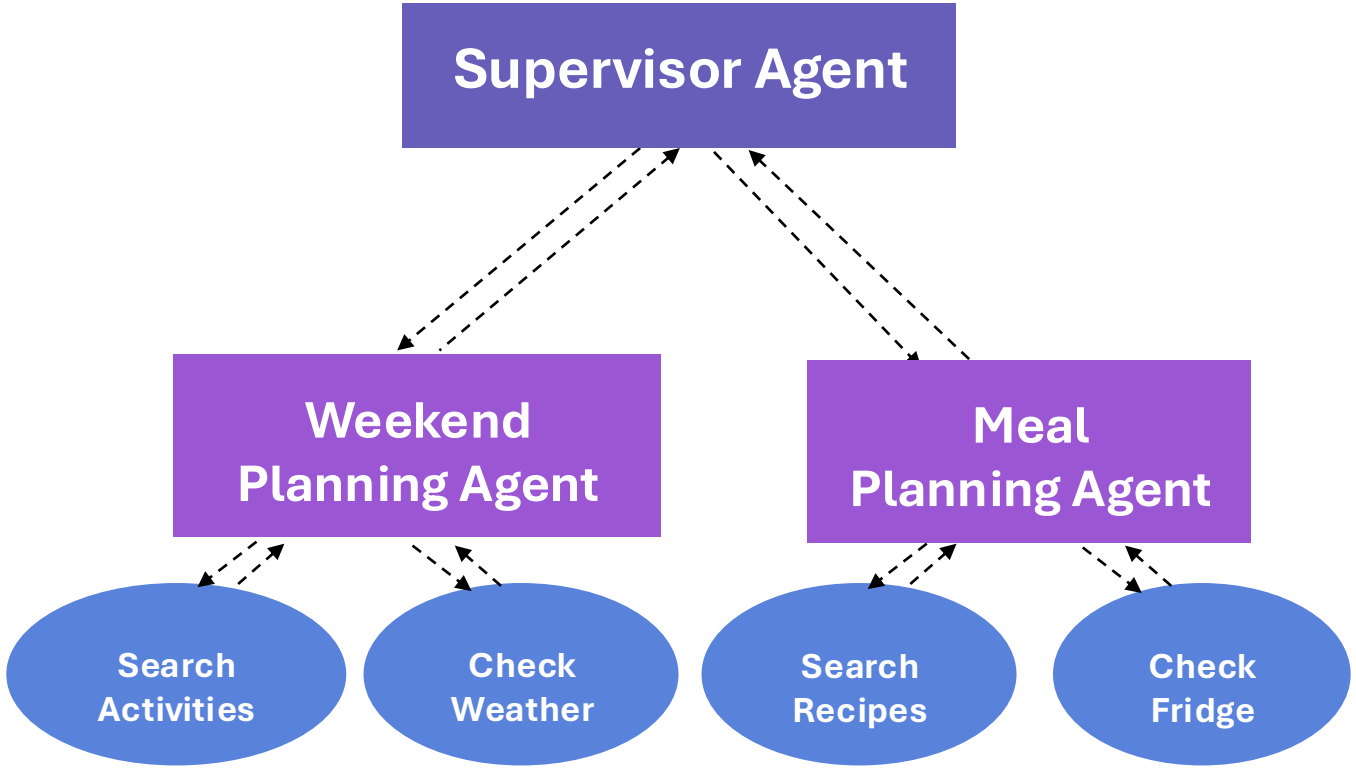
Supervisor agent with sub-agents

The **supervisor** agent decides which **specialist** agents should handle a task. Each agent may still have associated tools to call.

General architecture



Parenting helper agent



agent-framework

Supervisor agent with sub-agents

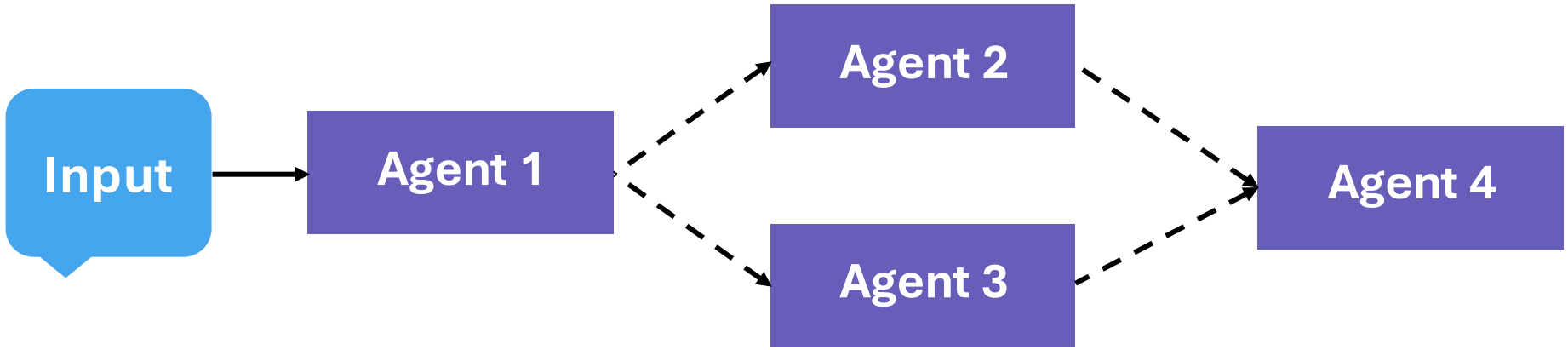
```
weekend_agent = Agent(client=client,
  instructions="You help users plan weekend activities based on weather",
  tools=[get_weather, get_activities, get_current_date])
meal_agent = Agent(client=client,
  instructions="You help users plan meals + choose the best recipes",
  tools=[find_recipes, check_fridge])

@tool
async def plan_weekend(query: str) -> str:
  response = await weekend_agent.run(query)
  return response.text
@tool
async def plan_meal(query: str) -> str:
  response = await meal_agent.run(query)
  return response.text

supervisor_agent = Agent(client=client,
  instructions="You're a supervisor managing a weekend planner & meal planner",
  tools=[plan_meal, plan_weekend])
```

Full example: agent_supervisor.py

Agent architecture:

Multi-agent workflows

For more complex multi-agent architectures, build workflows:



Workflows can include conditional edges, approvals, human-in-the-loop, and non-agent nodes as well.

Join week 2 of this series to learn how to build workflows with agent-framework!

Next steps

Register:
https://aka.ms/PythonAgents/series

Watch past recordings:
aka.ms/pythonagents/resources

Join office hours after each session in Discord:
aka.ms/pythonai/oh

🛠️ Feb 24: Building your first agent in Python
🧠 Feb 25: Adding context and memory to agents
📊 Feb 26: Monitoring and evaluating agents
🔀 Mar 3: Building your first AI-driven workflows
🕸️ Mar 4: Orchestrating advanced multi-agent workflows
🙋🏽‍♀️ Mar 5: Adding a human-in-the-loop to workflows

Appendix

Learn more

Documentation:

https://learn.microsoft.com/agent-framework/

Watch Ignite talks:

- AI powered automation & multi-agent orchestration in Azure AI Foundry
- Innovation Session: Your AI Apps and Agent Factory
- Multi-Agent Apps with Microsoft Agent Framework or LangGraph
- Build A2A and MCP Systems using SWE Agents and agent-framework
- AI builder's guide to agent development in Foundry Agent Service

Join weekly Office Hours:

https://github.com/microsoft/agent-framework/blob/main/COMMUNITY.md

Migration tips

MAF vs Semantic Kernel and AutoGen

	Semantic Kernel	AutoGen	Agent Framework
Focus	Stable SDK with enterprise connectors, workflows, and observability	Experimental multi-agent orchestration from research	Unified SDK combining innovation + enterprise readiness
Interop	Plugins, connectors, and support for MCP, A2A, OpenAPI	Tool integration supported; lacks standardized cross-runtime protocols	Built-in connectors, MCP + A2A + OpenAPI
Memory	Multiple vector store connectors and memory store abstraction (e.g. Azure SQL Elasticsearch, MongoDB)	Support for in-memory / buffer history + external vector store memory options (ChromaDB, Mem0, etc)	Pluggable memory across stores (first-party and third-party), persistent & adaptive memory stored with retrieval, hybrid appraoches
Orchestration	Deterministic + dynamic orchestration (Agent Framework, Process Framework)	Dynamic LLM orchestration (debate, reflection, facilitator/worker, group chat)	Deterministic + dynamic orchestration (Agent Orchestration, Workflow Orchestration)
Enterprise readiness	Telemetry, observability, compliance hooks	Minimal	Observability, approvals, CI/CD, long-running durability, hydration

https://devblogs.microsoft.com/foundry/introducing-microsoft-agent-framework-the-open-source-engine-for-agentic-ai-apps/

Autogen → Agent-framework

- AutoGen pioneered many orchestration patterns (GroupChat, GraphFlow, event-driven runtimes), which are now unified in Agent Framework under the **Workflow** abstraction that supports checkpointing, pause/resume, and human-in-the-loop flows.
- The **AssistantAgent** maps directly to the new **ChatAgent**, which is multi-turn by default and continues tool invocation until a result is ready.
- FunctionTool wrappers migrate to the **@ai_function** decorator, with automatic schema inference and support for hosted tools like code interpreter or web search.
- Most single-agent migrations require only light refactoring; multi-agent migrations benefit from the new **Workflow** model with stronger composability and durability.

Migration guide:
https://learn.microsoft.com/agent-framework/migration-guide/from-autogen/

Code samples (Python):
https://github.com/microsoft/agent-framework/tree/main/python/samples/autogen-migration

Semantic Kernel → Agent-framework

- Replace **Kernel** and **plugin** patterns with the **Agent** and **Tool** abstractions.
- Agents now manage threads natively, simplify invocation with RunAsync / RunStreamingAsync, and register tools inline without attributes or plugin wrappers.
- Existing vector store integrations (Azure AI Search, Postgres, Cosmos DB, Redis, Elasticsearch, etc.) continue to work through connectors.
- Plugins like Bing, Google, OpenAPI, and Microsoft Graph port directly as tools, often exposed via MCP or OpenAPI.

Migration guide:
https://learn.microsoft.com/agent-framework/migration-guide/from-semantic-kernel/

Code samples:
https://learn.microsoft.com/agent-framework/migration-guide/from-semantic-kernel/samples

Use GitHub Copilot for migration

Try out a prompt like:

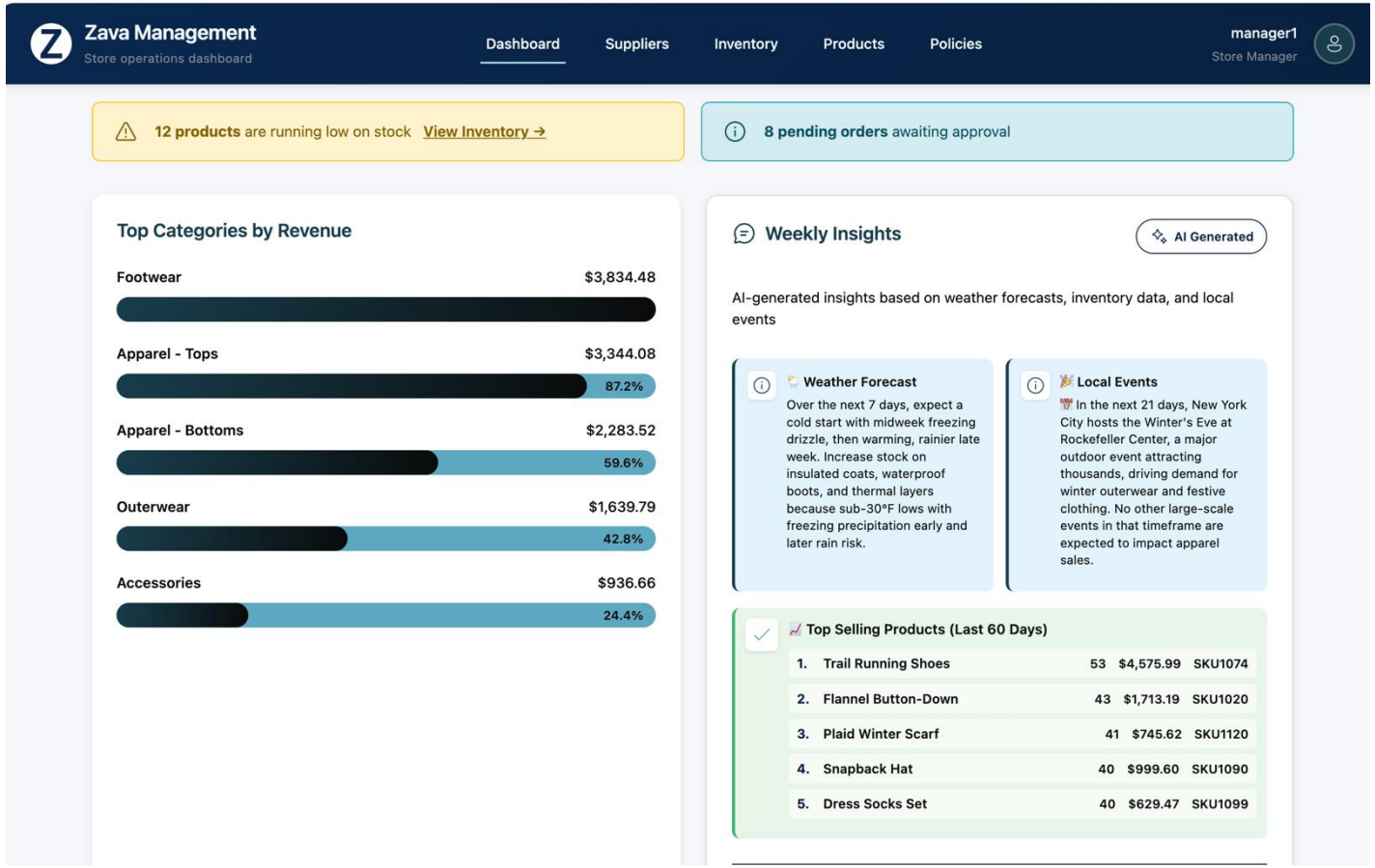
Port this code from AutoGen to Agent Framework, following the guide at https://learn.microsoft.com/en-us/agent-framework/migration-guide/from-autogen/

Find relevant code samples here if needed:

https://github.com/microsoft/agent-framework/tree/main/python/samples/autogen-migration

E2E sample

Sample: Agentic Shop



This app uses docker-compose to deploy 2 MCP servers, an API server (that runs agents), plus a frontend server.

```
agent = chat_client.create_agent(
  name="Retail Insights Analyzer",
  tools=[finance_mcp],
)
response = await agent.run(
  "Top 5 selling products for store 1",
  response_format=ProductsAgentResponse
)
products = response.value.products
```

https://github.com/tonybaloney/agentic-popup-shop